

Battery Degradation and State-of-Health Digital Twin

From executable electro-thermal scenarios to leakage-controlled measured SOH validation.

Electro-thermal

1RC Thevenin ECM, Coulomb SOC, heat balance

Aging

Calendar and cycle fade scenarios

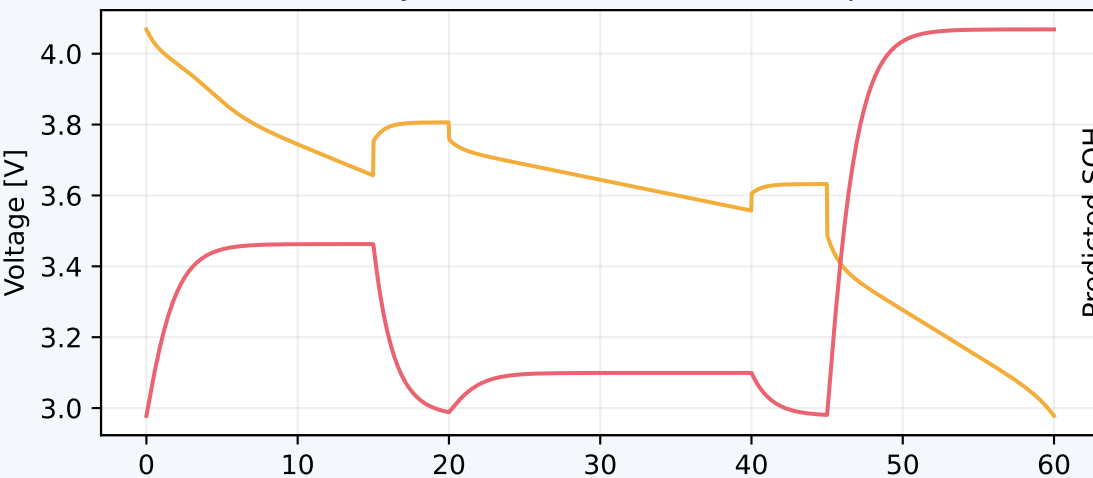
Measured evidence

478 measurements, six chemistries

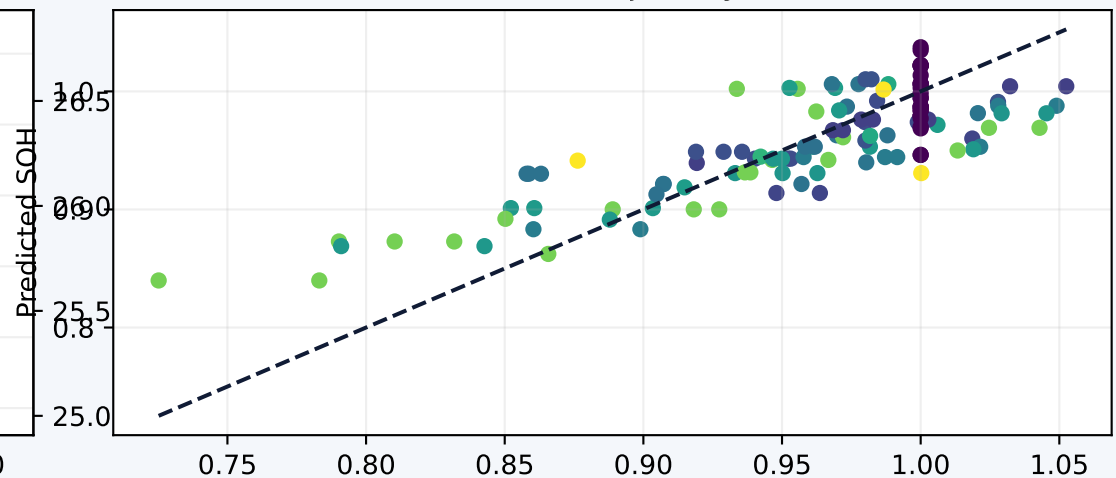
Research outputs

CSV, JSON, plots, notebooks, tests

Level 0 synthetic electro-thermal response



Level 1 held-out trajectory validation



478

Measured rows

27

Held-out groups

0.028

SOH MAE

0.044

Baseline MAE

3 passed

Tests

1 Define profile 2 Simulate physics 3 Validate schema 4 Calibrate by groups 5 Report held-out evidence